HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained automated test scripts in **Java** and **Python** to validate middleware components for in-vehicle infotainment systems, achieving a **92% test coverage** for core functionalities.
- Designed and implemented test scenarios to proactively identify potential software issues in complex distributed systems, reducing critical bugs found in later stages by **28%**.
- Collaborated with software engineers and product managers to translate product features and technical requirements into comprehensive test plans and procedures.
- Executed test scripts, analyzed test completion metrics, and generated detailed status reports for senior management, ensuring transparency on software quality and stability.
- Troubleshot and resolved technical and networking issues across the full technology stack, from application layers to underlying communication protocols, for enterprise-grade software.
- Contributed to the development and enhancement of automated testing methodologies and frameworks, focusing on reusable components and efficient test execution.
- Containerized test environments using **Docker** for consistent deployment across development and testing stages, integrating with **Jenkins** for automated build and test pipelines.

SKILLS

Testing Frameworks & Methodologies – Automated Testing, Test Automation Frameworks, Google Test, Unit Testing, Test Plans, Test Scripts, Traceability Matrices, Test Methodologies

Programming Languages – Java, Python, C/C++, SQL

Build & Configuration Management – CMake, Git

Cloud & CI/CD – AWS, Docker, Jenkins

Operating Systems & Networking – Linux, Windows, Networking, Communication Protocols

Software Design Principles – Object-Oriented Programming (OOP), Design Patterns

PROJECTS

Automated Test Framework for Embedded Systems

- Engineered a **Java**-based test automation framework to validate firmware updates for connected vehicle systems, integrating with **Google Test** for unit testing.
- Developed Python scripts to automate data generation and inject specific network conditions for comprehensive end-to-end testing scenarios.
- Implemented test execution tracking and reporting mechanisms, providing detailed logs and results for over **1,500 test cases** per release cycle.

Middleware Integration Testing Suite

- Built a suite of automated tests in **Python** to verify the integrity and performance of middleware services responsible for data ingestion and processing.
- Designed test cases to simulate high-volume traffic and error conditions, achieving a **35% reduction** in integration-related defects before production deployment.
- Configured test environments using **Docker** and managed dependencies with **CMake**, ensuring reproducible test runs across different development machines.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)